

Chapter 104 — Pathophysiology and Classification of Lower Urinary Tract Dysfunction

- The **lower urinary tract (LUT)** works as one interrelated unit whose adult purpose is **low-pressure storage with continence and complete voluntary expulsion, also at low pressure**.
- **Normal sensation**: a gradually rising, **deferrable sense of fullness** up to a volume that prevents frequency — without pain or **urgency** (a sudden compelling desire to void that is difficult to defer).
- The adult LUT is **under voluntary neural control** — unlike other autonomically innervated viscera, which are regulated solely by involuntary mechanisms.
- The micturition cycle is taught as **two discrete phases — filling/storage and emptying/voiding** — operating in an **on-off fashion**: inhibit voiding reflex + activate storage reflexes ⇌ inhibit storage reflexes + activate voiding reflex.

NORMAL LOWER URINARY TRACT FUNCTION: OVERVIEW

Two-Phase Concept of Function: Filling/Storage and Emptying/Voiding

FILLING/STORAGE VS EMPTYING/VOIDING — REQUIREMENTS

Filling/storage requires	Emptying/voiding requires
Accommodation of rising volumes at low detrusor pressure (normal compliance) with appropriate sensation	Coordinated bladder smooth-muscle contraction of adequate magnitude and duration
Outlet closed at rest and during rises in intra-abdominal pressure	Concomitant lowering of smooth and striated sphincter resistance (no functional obstruction)
No involuntary contractions — detrusor overactivity (DO)	No anatomic obstruction

SMOOTH VS STRIATED SPHINCTER

Smooth sphincter	Striated sphincter
Smooth muscle of bladder neck and proximal urethra	Intrinsic (intramural) — rhabdosphincter : striated muscle in outer wall of proximal urethra, both sexes Extrinsic (extramural) : skeletal muscle around membranous urethra (male) / mid-segment (female) = classic external urethral sphincter
Physiological, not anatomical , sphincter	Anatomical structures
Not under voluntary control	Extramural portion under voluntary control

Bladder Response During Filling

- Physiological filling → **almost no pressure change**; once the wall unfolds, **high compliance (Δ volume/ Δ pressure)** reflects **elastic and viscoelastic properties**.
 - **Elasticity** — wall constituents stretch **without any rise in tension**.
 - **Viscoelasticity** — stretch raises tension, which then decays (“**stress relaxation**”) when filling slows; set by the **extracellular matrix** (elastic fibres, collagen).
- Smooth muscle may make **continuous asynchronous length adjustments** (never synchronous, pressure-raising activity); **actomyosin cross-bridge regulation** may set compliance. The bladder appears “**relaxed**”; the **urothelium expands yet keeps its barrier**.

- A possible **non-neurogenic active storage component**: an unidentified **urothelial relaxing factor**; urothelial **nitric oxide (NO)** may also inhibit afferents.
- **Passive compliance = viscoelastic stroma (collagen, elastin) + urodynamically non-contractile detrusor.**
- **Klevmark**: the pressure rise on cystometry is an **artefact of supraphysiological filling**; physiologically, **none until capacity.**
- **↑ collagen → ↓ compliance**: chronic inflammation, **bladder outlet obstruction (BOO)**, neurological decentralisation, injury; **hypertrophic detrusor** is less elastic and makes more collagen. **Once established, collagen-driven low compliance is generally drug-unresponsive.**
- **Neural storage reflexes** (clear in animals, indirect in humans): mainly **sympathetic inhibition of cholinergic ganglionic transmission**, plus:
 - **α₁-adrenergic stimulation at the smooth sphincter → ↑ outlet resistance.**
 - **β₃-adrenergic (inhibitory) stimulation of the bladder body → ↓ wall tension.**
 - **Pudendal afferents from the striated sphincter directly inhibit sacral detrusor motor neurons.**
 - Other inhibitory transmitters: **glycine, γ-aminobutyric acid, opioids, purines, noradrenergic system.**
 - Distension releases **urothelial factors** (acetylcholine, adenosine triphosphate, NO, prostaglandins, peptides) that modulate contractility.
- **Current urodynamics cannot measure true detrusor wall tension.**

Outlet Response During Filling

- **Proximal urethral pressure rises gradually during filling**, tracking **efferent pudendal firing and sphincter electromyographic activity** — the **guarding reflex** (spinal somatic). A **sympathetic smooth-sphincter contribution** is plausible but **very hard to prove.**
- Urethral pressure = active muscle **plus passive elastic, collagenous and vascular components**; wall tension compresses a **submucosal “filler material”** — **the softer, the less pressure continence needs.**
- **Mucosal seal mechanism**: the lumen must close watertight — a greased rubber tube closes with less pressure; **scarred or atrophic mucosa** seals poorly.
- **Oestrogen**: shared **urogenital sinus** origin, urethral receptors, menopausal **urinary incontinence (UI)** — yet **oestrogen for postmenopausal UI remains controversial.**

Voiding With a Normal Bladder Contraction

- **Trigger**: rising intravesical pressure producing the **sensation of distension.**
- Parasympathetic outflow (**pelvic nerve**) is **sacral**, but the **coordinating centre is the rostral brainstem**, modulated by **cortex.**
- Sequence: **inhibit somatic efferents to striated sphincter + all spinal sympathetic storage reflexes → parasympathetic pelvic-nerve efferents → coordinated contraction of bulk detrusor.**
- **Outlet resistance falls** by: continence reflexes ceasing; **active smooth-sphincter relaxation via a non-adrenergic non-cholinergic (NANC) mechanism, probably NO**; **funnelling** — longitudinal smooth-muscle **continuity** of base and proximal urethra shortens and widens the urethra.
- Reflexes from contraction and **urethral flow** reinforce emptying; **supraspinal inputs** give conscious control.

Urinary Continence During Abdominal Pressure Increases

- Why does coughing not open the outlet? (1) **No coordinated contraction is triggered** — **raised intravesical pressure ≠ emptying ability**; (2) urethral entry needs a **neurally mediated contraction plus bladder neck conformational change.**

- Key protection: **at least equal pressure transmission to the proximal urethra (and female mid-urethra) — Enhorning (1961)**. Its failure invariably accompanies effort-related UI in both sexes.
- **Tanagho (1978)**: closure pressure rises with abdominal pressure — first direct evidence of an **active reflex component** beyond passive transmission.

Sensory Aspects

AFFERENT PATHWAYS OF THE LOWER URINARY TRACT

Nerve	Carries	Relevance
Pelvic (via dorsal root ganglia → sacral cord)	Most bladder/urethral afferents: tension, volume and nociceptive receptors in serosa, muscle, urothelium, suburothelium	Most important for initiating and maintaining micturition
Hypogastric	Some bladder/urethral afferents	Minor route
Pudendal	Striated sphincter and pelvic floor	Somatic afferents

- Normal filling gives **fullness, not urgency or pain**, and initiates voiding reflexes. In **urologic chronic pelvic pain syndrome (UCPPS)**, nociception disturbs **brain-level sensorimotor regulation of storage**.

Concept — *Storage is **mostly passive** (wall mechanics plus absent parasympathetic drive) with reflex outlet tightening; voiding is **active and brainstem-coordinated**, and continence under stress depends on the outlet receiving **at least the pressure the bladder receives**.*

MICTURITION CYCLE: SIMPLIFICATION AND OVERVIEW

THE MICTURITION CYCLE IN BRIEF

Filling/storage	Emptying/voiding
Passive accommodation, no parasympathetic drive	Voluntary or involuntary; somatic + sympathetic reflexes off, parasympathetics on ; centre rostral brainstem
Guarding reflex ↑ outlet resistance	First outlet resistance falls (reflexes cease ± relaxing factor)
Some species: sympathetic reflex — ↑ smooth-sphincter tension, ganglionic inhibition, ↓ bladder-body tension	Then coordinated detrusor contraction with funnelling
Stress continence: intrinsic outlet competence + pressure transmission ratio + reflex sphincter activity	Reflex amplification → complete emptying if no anatomic or functional obstruction

ABNORMALITIES OF FILLING/STORAGE AND EMPTYING/VOIDING: OVERVIEW OF PATHOPHYSIOLOGY

- Excluding psychological causes, every failure to fill, store or empty is due to **the bladder, the outlet, or both** — the basis of the **Wein functional classification**.
- Some patients have **combined storage and emptying failure** — e.g. **suprasacral spinal cord injury: DO + striated sphincter dyssynergia**; or **DO during filling with detrusor underactivity during emptying**.
- Failure is usually **relative**. The scheme expands (Box 104.2), but the **simple form avoids argument when mechanism is uncertain**, and files **every test** (Table 104.1) and **treatment** (Boxes 104.3–104.4) by phase and site.

BOX 104.1 SIMPLE FUNCTIONAL CLASSIFICATION OF LOWER URINARY TRACT DYSFUNCTION

- **Failure to store** — because of the bladder / because of the outlet.
- **Failure to empty** — because of the bladder / because of the outlet.

Courtesy Dr. Alan J. Wein.

BOX 104.2 EXPANDED FUNCTIONAL CLASSIFICATION OF LOWER URINARY TRACT DYSFUNCTION

Category	Because of the bladder	Because of the outlet
I. Failure to store	1. Overactivity — (a) involuntary contractions (DO) : neurological disease/injury/degeneration; BOO; ↑ afferent input or sensitivity (inflammation, ↑ transmitter release, ↑ transmitter sensitivity); ↓ inhibitory pelvic floor activity; idiopathic. (b) Decreased compliance : neurological; fibrosis; detrusor hypertrophy; idiopathic. (c) Combination. 2. Hypersensitivity — inflammatory/infectious; neurological; ↑ transmitter release or sensitivity; psychological; idiopathic. 3. Underactivity (retention with overflow incontinence). 4. Combination.	1. Genuine stress UI — lack of suburethral support; pelvic floor laxity, hypermobility. 2. Intrinsic sphincter deficiency — neurological; fibrosis. 3. Combination of both.
C. Combination (bladder + outlet) · D. Fistula		
II. Failure to empty	Underactivity : neurogenic (includes iatrogenic); myogenic; psychogenic; idiopathic.	1. Anatomic — prostatic obstruction; bladder neck contracture; urethral stricture; urethral compression/fibrosis. 2. Functional — striated sphincter dyssynergia (neurogenic) ; smooth sphincter dyssynergia/dysfunction (bladder neck dysfunction) ; dysfunctional voiding (non-neurogenic) .
C. Combination		

Courtesy Dr. Alan J. Wein.

- **Pattern**: a 2 × 2 grid (phase × site); **overactivity = phasic DO or low compliance**; fistula is the only storage failure outside bladder and outlet.
- **Trap**: **bladder underactivity appears under failure to store** (overflow incontinence) as well as failure to empty.

TABLE 104.1 URODYNAMICS SIMPLIFIED

Phase	Bladder	Outlet
Filling/storage	Total bladder (P_{ves}) and detrusor (P_{det}) pressures on filling cystometrogram (FCMG) ; detrusor leak point pressure (DLPP)	Urethral pressure profilometry (UPP) ; Valsalva leak point pressure (VLPP) ; fluoroscopy of outlet during filling
Emptying	P_{ves} , P_{det} on voiding cystometrogram (VCMG)	Fluoroscopy of outlet during emptying; micturitional UPP ; electromyography (EMG) of periurethral striated muscle
Emptying — integrates both	Flowmetry and residual urine	

- **Trap**: **flow rate and post-void residual cannot localise the fault** — they integrate bladder *and* outlet; a poor flow may be weak detrusor or obstruction.

- **Pair: DLPP = bladder (storage) vs VLPP = outlet (storage).**

Filling and Storage Failure

- Results from **bladder overactivity** (involuntary contraction and/or ↓ compliance), ↓ outlet resistance, heightened or altered sensation, or a combination.

Bladder Overactivity

- Expressed as **phasic involuntary contractions, low compliance, or both.**
- Involuntary contractions: **neurological disease, BOO, stress urinary incontinence (SUI)** (urine entering the proximal urethra may trigger one), **ageing**, or **idiopathic**; **micromotions** may drive afferents to DO. Also ↑ **afferent input** (inflammation) or ↑ **sensitivity**: urothelial **excitatory transmitters** → DO or heightened sensation.
- **Urgency urinary incontinence (UUI) → assume DO. Urgency alone suggests DO, often not shown on urodynamic studies (UDS); urodynamic DO may be asymptomatic.**
- **Overactive bladder (OAB)** — International Continence Society (ICS): **urgency ± UUI, usually with frequency and nocturia.** Possible mechanisms:
 1. ↓ suprapontine inhibition
 2. Damaged spinal axonal paths
 3. Damaged peripheral axonal paths
 4. Loss of peripheral inhibition
 5. Enhanced excitatory transmission in the micturition reflex
 6. ↑ LUT afferent input
 7. Idiopathic
- ↓ **pelvic floor stimulation** may contribute; pelvic floor contraction engages the **frontal-cortex voluntary inhibition reflex** — basis of **bladder training**.

Filling and Storage Failure Resulting From Altered Sensation

- Storage failure **without overactivity** — inflammation, irritation, hypersensitivity, pain (chemical, psychological, idiopathic); classically **interstitial cystitis/bladder pain syndrome (IC/BPS)**. ↑ afferent activity gives **DO, urgency without DO, premature fullness alone, or pain on filling.**

Filling and Storage Failure From Remodeling of the Bladder Leading to Decreased Compliance

- ↓ compliance: **neurological injury, usually sacral or infrasacral**, or anything destroying wall elasticity, e.g. **collagen from long-standing BOO.**

Outlet Underactivity

- ↓ outlet resistance from damage to **sphincter innervation or structure**, or to **female outlet support**: neurological disease, surgical/mechanical trauma, ageing.

GENUINE STRESS INCONTINENCE (HYPERMOBILITY) VS INTRINSIC SPHINCTER DEFICIENCY (ISD)

Genuine stress incontinence / hypermobility	Intrinsic sphincter deficiency (formerly “type III”)
Poor pelvic support ; outlet competent at rest , fails only with ↑ abdominal pressure	Non- or poorly functional bladder neck/proximal urethra at rest
Rotational descent of bladder neck and urethra (classic); or descent without rotation (shortens/widens); or posterior wall sheared open, anterior fixed	Overt form: bladder neck open at rest, low abdominal leak point pressure and low urethral closure pressure

Genuine stress incontinence / hypermobility	Intrinsic sphincter deficiency (formerly “type III”)
Hypermobility alone is not diagnostic — common in continent females; needs demonstrated UI	Usually from prior surgery, trauma with scarring, or neurological lesion
Treated by correcting hypermobility	Hypermobility-only surgery has high failure ; needs coaptation/compression

Contemporary view: most female effort UI = varying mix of support loss and ISD. Incontinence can be due to ISD alone, never to hypermobility alone — some ISD must exist.

- **DeLancey’s hammock hypothesis:** SUI with hypermobility requires an **unstable suburethral supportive layer**. Firm layer → abdominal pressure compresses the urethra; **lax/mobile layer → compression fails**.
- SUI = pelvic floor **muscle, nerve or connective tissue damage**; the female urethra is supported by **levator ani via anterior vaginal wall endopelvic fascia**. **Loss of bladder neck closure** causes UI even with normal support.
- Continence was once placed solely at the bladder neck, but in about **half of continent females urine enters the urethra on straining**; their **continence point (highest pressure transmission) is the mid-urethra**.
- **Urethral instability:** episodic urethral pressure falls without bladder or abdominal change; **rarely demonstrable**; probably the urethral half of a reflex in a bladder that **does not measurably contract**.
- **Males:** categories theoretically parallel, but **male sphincteric incontinence is not associated with hypermobility — it resembles female intrinsic sphincter dysfunction**. Almost no data on male urethral instability.
- **Treatment of storage failure:** inhibit contractility, ↓ sensory output, mechanically ↑ capacity, or ↑ outlet resistance (continuously or only during abdominal pressure rises).

BOX 104.3 FUNCTIONAL CATEGORISATION OF THERAPY TO FACILITATE URINE STORAGE/BLADDER FILLING

Target	Modality	Options
I. Bladder (↓ intravesical pressure, inhibit contractility, ↑ capacity)	Behavioural	Education; bladder training; timed/prompted voiding; fluid restriction; pelvic floor physiotherapy ± biofeedback
	Pharmacological (oral, intravesical, intradetrusor)	Antimuscarinics ; mixed-action drugs; β-adrenergic agonists ; onabotulinumtoxinA ; calcium antagonists; potassium channel openers; prostaglandin inhibitors; α-adrenergic antagonists; tricyclic antidepressants; serotonin–noradrenaline reuptake inhibitors; dimethyl sulfoxide; polysynaptic inhibitors; capsaicin, resiniferatoxin and similar
	Mechanical	Bladder overdistension
	Electrical	Sacral neuromodulation ; posterior tibial and other peripheral nerve stimulation
	Acupuncture	Acupuncture, electroacupuncture
	Denervation	Very central (subarachnoid block); less central (sacral/selective sacral rhizotomy); peripheral motor and/or sensory
II. Outlet (↑ resistance)	Surgical	Augmentation cystoplasty (auto-augmentation, bowel, tissue engineering)
	Behavioural, electrical	As I.A; electrical stimulation
	Pharmacological	α-adrenergic agonists ; tricyclics, serotonin–noradrenaline reuptake inhibitors; β-adrenergic antagonists, agonists

Target	Modality	Options
	Devices, bulking	Vaginal/perineal occlusive or supportive devices; urethral plugs; periurethral bulking (synthetics, tissue engineering)
	Female surgery	Retropubic suspension, slings, mid-urethral tapes — each ± prolapse repair
	Other surgery	Perineal sling (male); artificial urinary sphincter ; myoplasty; bladder outlet closure
III. Circumvent	—	Absorbent products; external collecting devices; antidiuretic hormone-like agents; short-acting diuretics; intermittent or continuous catheterisation; urinary diversion

Courtesy Dr. Alan J. Wein.

- **Pattern:** the same adrenergic receptor is used in opposite directions by site — α -agonist tightens the outlet (storage), α -antagonist relaxes it (emptying); β -agonist relaxes the bladder (storage).
- **Trap:** catheterisation and diuretics/antidiuretics are “circumventing” — they treat the consequence, not the bladder or outlet.

Emptying and Voiding Failure

- Results from ↓ **bladder contractility** (magnitude, coordination or duration), ↑ **outlet resistance**, or both.

Bladder Underactivity

- Temporary or permanent failure of the **neuromuscular mechanisms that initiate and sustain a detrusor contraction**.
- In the neurologically normal: **reflex inhibition from ↑ pelvic/perineal afferent input**, or **psychogenic**.
- **Iatrogenic:** colorectal, urological and gynaecological surgery.
- **Myogenic:** overdistension; central or peripheral drugs; severe infection; fibrosis.
- **Diabetes mellitus:** combined **neuromuscular and vascular** mechanism.

Outlet Overactivity or Obstruction

- **ICS bladder outflow obstruction** = **reduced flow and/or raised post-void residual (PVR) with raised detrusor pressure; much commoner in males; usually anatomic**, sometimes **functional** (sphincter fails to relax or contracts).
- **Hallmark of BOO, whatever the cause: high detrusor pressure producing low flow.**

FUNCTIONAL OUTLET OBSTRUCTION — THE ENTITIES

Entity	Level	Setting	Key feature
Striated sphincter dyssynergia	Striated sphincter	Neurological disease/injury	Common cause of functional obstruction
True smooth sphincter dyssynergia	Bladder neck/proximal urethra	Autonomic dysreflexia	Otherwise unusual
Bladder neck dysfunction (primary bladder neck obstruction, dysfunctional bladder neck)	Anatomically normal bladder neck	Young and middle-aged	Incomplete opening during voiding; uncommon
Dysfunctional voiding	Striated sphincter	Non-neurogenic	Listed in Box 104.2

ANATOMIC OUTLET OBSTRUCTION — MALE VS FEMALE

Males	Females
Prostatic enlargement; bladder neck contracture; urethral stricture	Pelvic organ prolapse; urethral diverticulum; compression or fibrosis after incontinence surgery

- **Treatment:** ↑ detrusor pressure, facilitate the reflex, ↓ outlet resistance. **Intervene promptly** — chronic obstruction causes **hypertrophy, collagen, diverticula, DO** and **neural change**; failing all else, **catheterisation** circumvents it.

BOX 104.4 FUNCTIONAL CATEGORISATION OF THERAPY TO FACILITATE BLADDER EMPTYING AND VOIDING

Target	Modality	Options
I. Bladder (↑ intravesical pressure, augment contractility)	Mechanical	External compression, Valsalva
	Reflex promotion	Trigger zones/manoeuvres; bladder “training”; tidal drainage
	Pharmacological (oral, intravesical)	Parasympathomimetics ; prostaglandins; blockers of inhibition (α-adrenergic antagonists, opioid antagonists)
	Electrical	Bladder or spinal cord; nerve roots; intravesical (transurethral); neuromodulation
	Surgical	Reduction cystoplasty; bladder myoplasty (muscle wrap); tissue engineering
II.A Anatomic obstruction	Pharmacological (↓ prostate size/tone)	α-adrenergic antagonists; 5α-reductase inhibitors ; luteinising hormone–releasing hormone agonists/antagonists; antiandrogens
	Procedural	Prostatectomy/prostatotomy (diathermy, heat, laser, stapling); bladder neck incision/resection; stricture repair/dilation; intraurethral devices; balloon dilation; prolapse correction (pessary or surgery)
II.B Smooth sphincter	Drug / procedure	α-antagonists ; β-agonists; onabotulinumtoxinA; transurethral resection/incision; reconstruction
II.C Striated sphincter	Drug / procedure	Behavioural ± biofeedback; benzodiazepines, baclofen, dantrolene , α-antagonists, onabotulinumtoxinA ; surgical sphincterotomy; pudendal nerve interruption
II.D Circumvent	—	Intermittent or continuous catheterisation; urinary diversion (conduit or reservoir)

Courtesy Dr. Alan J. Wein.

- **Pattern:** outlet therapy is organised **by level** — **anatomic site, smooth sphincter, striated sphincter**; skeletal-muscle relaxants act only at the striated level.
- **Trap:** **onabotulinumtoxinA** appears in **both boxes** — intradetrusor for storage, sphincteric for emptying; the site decides the effect.

Concept — Every dysfunction is **a bladder or an outlet failing to store or to empty**; because therapies are filed on the same two axes, **naming the failure names the treatment family**.

ABNORMALITIES OF PAIN: CHRONIC PELVIC PAIN SYNDROME

- Hallmark of IC/BPS is **pain**, but **frequency and urgency** are very common.
- Classic urgency is “a **compelling need to urinate which is difficult to postpone**”; IC/BPS urgency is **qualitatively different** — **driven by pain, pressure or discomfort that worsens with filling**.
- May present with a single symptom, then **progresses to the full constellation** (pain, nocturia, urgency/frequency) **over about two years**.

Central Sensitization

- **Definition:** ↑ responsiveness of **central nociceptive neurons** to normal or subthreshold afferent input. Implicated in fibromyalgia, chronic fatigue syndrome, headache, **irritable bowel syndrome (IBS)** — and recently theorised in **OAB urgency and frequency**.

BLADDER AFFERENTS — Aδ VS C FIBRES

Aδ fibres	C fibres
Small-calibre, myelinated	Small-calibre, unmyelinated
Predominant normal bladder afferent	Normally silent
Respond to pressure and volume ; continuously gauge wall distension	Activated only by extreme distension, cold, heat, chemical irritation ; become hyperactive to low-intensity input in OAB and central sensitisation models

- Sensitised pathways overlap with other pelvic viscera — “**pelvic organ crosstalk**” (bladder–bowel) — explaining **co-occurrence of OAB and IBS**.

Abnormalities of Nocturia

- **Nocturia:** waking **one or more times** during intended sleep to void; variable bother; from LUT dysfunction and/or other medical conditions.

POLYURIA VS NOCTURNAL POLYURIA

	Polyuria	Nocturnal polyuria
Definition	Excess 24-hour output: >40 mL/kg/24 h (≈ 2.8 L for 70 kg)	Isolated excess night-time output relative to day
Measure	24-hour volume	Nocturnal Polyuria Index (NPI) = nocturnal volume (including first morning void) $\div$ 24-hour volume
ICS threshold	—	>20% (younger); >33% (elderly)
Causes	↑ intake; diabetes mellitus ; central or nephrogenic diabetes insipidus ; primary polydipsia	Loss of nocturnal vasopressin rise with age; obstructive sleep apnoea (OSA)

- **Vasopressin** increases tubular water reabsorption; it **peaks at night in young adults** but the **nocturnal rise is lost in older adults**.
- **OSA** causes nocturia by (1) **repeated awakening** and (2) **↑ atrial natriuretic peptide (ANP) → vasopressin suppression → nocturnal polyuria**.

Neurogenic Lower Urinary Tract Dysfunction

- **ICS definition of NLUTD:** abnormal or difficult bladder/urethral ($\pm$ prostatic) function **in the context of a clinically confirmed relevant neurological disorder**.
- Symptoms are related to **storage, voiding and post-voiding**; diagnoses combine **symptoms with signs**, often urodynamic.

NEUROGENIC LOWER URINARY TRACT DYSFUNCTION BY LESION LEVEL

Lesion	Site	Detrusor	Sphincter / emptying
Suprapontine	Above pons (forebrain, midbrain)	Reflex detrusor contraction , impaired cerebral regulation	Usually synergic — complete emptying

Lesion	Site	Detrusor	Sphincter / emptying
Suprasacral spinal	Spinal cord between pons and sacral cord	DO	± detrusor-sphincter dyssynergia (DSD) → often raised PVR
Sacral spinal	Sacral cord	Acontractile	Usually impaired sphincter activity

- **Trap: DSD needs an intact sacral arc below a disconnected pontine centre** — hence suprasacral, not suprapontine.

Concept — *Pain, nocturia and neurogenic dysfunction share the chapter's logic: **locate the failure in the pathway** — afferent sensitisation, urine production, or lesion level — and the phenotype follows.*

CLASSIFICATION SYSTEMS FOR VOIDING DYSFUNCTION

- A classification conveys **the clinical essence in a few words**; the ideal implies **urodynamic conclusions, expected symptoms and lesion site**, yielding a **treatment “menu”**. Most were built for **neurogenic dysfunction**; the ideal covers **all**.

THE FOUR CLASSIFICATION SYSTEMS COMPARED

	By function	By symptoms	By signs	By urodynamics
Basis	Storage vs emptying × bladder vs outlet	Patient's/carer's subjective report	Objective findings — diaries, examination, pad test	UDS findings , storage and voiding separately
Diagnosis needed?	No — exact diagnosis or UDS not required to treat	Cannot alone make a diagnosis ; directs testing	Supports diagnosis	Most precise

Classification by Function

- Weighs symptoms and UDS together to set the **timing and order of treatment**.
- **Combination deficits must be recognised**. Classic example — **T10 paraplegia after spinal shock: storage failure (DO) + emptying failure (striated sphincter dyssynergia)**. Designate one deficit primary, treat it, address the other secondarily:
 - Circumvent emptying (**external collection device**) → treat as **primary storage failure**; or
 - Abolish DO (drugs or surgery) → treat resulting emptying failure with **clean intermittent catheterisation**.
- Other combinations: impaired contractility or overactivity with sphincter dysfunction; BOO with DO; BOO with sphincter malfunction; **DO with impaired contractility**.

Classification by Symptoms

ICS LOWER URINARY TRACT SYMPTOMS BY PHASE

Phase	Symptoms
Storage	Daytime frequency (voiding too often by day); nocturia (waking one or more times to void, ± enuresis); urgency ± incontinence; UI subtyped — UUI (leak accompanied or preceded by urgency), SUI (leak on exertion), continuous ; bladder sensation — normal, increased, reduced, absent, non-specific
Voiding	Slow stream (vs previous or others); splitting ; intermittency (starts and stops); hesitancy (difficulty initiating); straining (effort to initiate or maintain); terminal dribbling (prolonged final part)

Phase	Symptoms
Post-micturition	Incomplete emptying; post-micturition dribble (involuntary loss immediately after finishing)

- **Trap: terminal dribbling is a voiding symptom; post-micturition dribble is post-micturition** — the difference is whether voiding has finished.

Classification by Signs

- Examination shows **SUI** (urethral leak synchronous with effort) and **anterior vaginal wall prolapse**, which may accompany UI or **mask it by kinking the urethra**; **pelvic floor function** is graded vaginally or rectally.

BOX 104.5 SIGNS SUGGESTIVE OF LOWER URINARY TRACT DYSFUNCTION AS DEFINED BY THE INTERNATIONAL CONTINENCE SOCIETY

- **Frequency–volume chart / bladder diary**: daytime frequency; nocturia; 24-hour frequency; 24-hour production; polyuria; nocturnal urine volume; nocturnal polyuria; **maximum voided volume**.
- **Examination**: UI (leak **visualised**); stress incontinence; **extraurethral incontinence**; uncategorised incontinence; **pelvic organ prolapse** (anterior, apical, posterior); **pelvic floor muscle function** (strong, weak, absent).
- **Pad testing**: quantifies urine lost per episode.

Classification by Urodynamics

- Storage and voiding described **separately**, each for **bladder and urethra**. **DO**: involuntary filling contractions, spontaneous or provoked — **neurogenic** with a relevant neurological condition, else **idiopathic**.
- **Normal urethral storage function: positive urethral closure pressure** (urethral minus bladder pressure) even with abdominal pressure rises. **Incompetent**: leakage **without detrusor contraction** — genuine stress incontinence, ISD, both, or an involuntary urethral pressure fall.

BLADDER CAPACITY — THREE DEFINITIONS

Term	Definition
Cystometric capacity	Volume at end of filling when “ permission to void ” is given
Maximum cystometric capacity	Volume at which the patient can no longer delay micturition
Maximum anaesthetic capacity	Volume filled under anaesthesia

VOIDING-PHASE URODYNAMIC TERMS

Term	Definition
Normal detrusor activity	Voluntarily initiated, sustained contraction → complete emptying in normal time
Detrusor underactivity	Contraction of inadequate magnitude and/or duration
Acontractile detrusor	No demonstrable contraction on testing
Normal urethral function	Opens and stays continuously relaxed — emptying at normal pressure
Abnormal urethral function	Mechanical obstruction or impaired relaxation
Detrusor sphincter dyssynergia	Detrusor contraction concurrent with involuntary urethral/periurethral striated muscle contraction

Term	Definition
Non-relaxing urethral sphincter obstruction	Usually with a neurological lesion ; non-relaxing obstructing urethra → reduced flow

- “**Neurogenic**” is prefixed to these terms when a relevant neurological disorder exists.

BOX 104.6 INTERNATIONAL CONTINENCE SOCIETY CLASSIFICATION

Domain	Storage phase	Voiding phase
Detrusor	Normal; DO — phasic or terminal, ± incontinence, neurogenic or idiopathic	Normal; underactive ; acontractile
Sensation	Normal; increased/early desire; reduced; absent; non-specific; bladder pain	—
Capacity	Normal; high; low	—
Compliance	Normal; high; low	—
Urethra	Normal closure; incompetent closure	Normal; abnormal — BOO , dysfunctional voiding , DSD , non-relaxing urethral sphincter obstruction

Modified from Abrams P, Cardozo L, Fall M, et al.: Standardisation Sub-committee of the International Continence Society, Neurourol Urodyn 21(2):167–178, 2002.

WORKED EXAMPLES — URODYNAMIC CLASSIFICATION

	T10 paraplegia (after spinal shock)	Stroke with UII
Detrusor (storage)	Neurogenic DO	Neurogenic DO
Sensation	Absent	Normal
Capacity / compliance	Low / normal	Low / normal
Urethra (storage)	—	Normal closure
Voiding	Detrusor sphincter dyssynergia	Normal detrusor and urethral function (if no anatomic obstruction)

Concept — *The four systems are **one functional frame viewed through progressively more objective lenses** — patient report, observed sign, urodynamic measurement; the paraplegic versus stroke contrast is **synergy**, which depends on whether the pons still coordinates the sacral arc.*